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University Place, Glasgow, G12 8QQ, UK

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**RESEARCH INTERESTS**

**Low-dimensional topology**, in particular braids, knots and links, cobordisms, knot concordance, knotted surfaces, Khovanov and Heegaard Floer homology, 3- and 4-manifolds.

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**EMPLOYMENT**

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 12/2025 – 11/2027 | <b>Postdoctoral fellow</b> , <a href="#">University of Glasgow</a> , Glasgow, UK.<br>Supported by SNSF Postdoc.Mobility fellowship 230329. |
| 10/2023 – 11/2025 | <b>Postdoctoral fellow</b> , <a href="#">Max Planck Institute for Mathematics</a> , Bonn, Germany.<br>Mentor: Arunima Ray.                 |
| 07/2019 – 07/2023 | <b>Scientific assistant</b> , <a href="#">ETH Zurich</a> , Zurich, Switzerland.                                                            |
| 10/2018 – 04/2019 | <b>Teaching assistant</b> , <a href="#">Ulm University</a> , Ulm, Germany.                                                                 |
| 08/2017 – 05/2018 | <b>Teaching assistant</b> , <a href="#">Syracuse University</a> , Syracuse, NY, USA.                                                       |
| 10/2013 – 07/2017 | <b>Teaching assistant</b> , <a href="#">Ulm University</a> , Ulm, Germany.                                                                 |

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**EDUCATION**

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07/2019 – 06/2023 | <b>Doctor of Science</b> in Mathematics, <a href="#">ETH Zurich</a> , Zurich, Switzerland.<br>Thesis: <i>On notions of braid positivity and knot concordance</i> .<br>Advisor: Peter Feller.                                |
| 08/2016 – 02/2019 | <b>Master of Science</b> in Mathematics (1.0, with distinction), <a href="#">Ulm University</a> , Ulm, Germany. Minor: Computer Science.<br>Thesis: <i>Massey products and linking numbers</i> . Supervisor: Stefan Wewers. |
| 08/2017 – 05/2018 | <b>Master of Science</b> in Mathematics (GPA 4.0), <a href="#">Syracuse University</a> , Syracuse, NY, USA.                                                                                                                 |
| 10/2012 – 08/2016 | <b>Bachelor of Science</b> in Mathematics (1.5), <a href="#">Ulm University</a> , Ulm, Germany.<br>Minor: Physics.<br>Thesis: <i>Das Existenzproblem von Hurwitz</i> . Supervisor: Stefan Wewers.                           |
| 09/2014 – 02/2015 | <b>Visiting semester</b> , <a href="#">ETH Zurich</a> , Zurich, Switzerland.                                                                                                                                                |

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**SELECTED AWARDS**

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|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 12/2025 – 11/2027 | <b>Swiss National Science Foundation Postdoc.Mobility fellowship</b> .<br><i>Project 230329: Concordance of braid positive knots</i> . |
| 2017              | <b>Fulbright Travel Grant</b> .                                                                                                        |
| 2014 – 2017       | <b>Talanx Scholarship</b> .                                                                                                            |
| 2013 – 2014       | <b>Deutschlandstipendium</b> Scholarship (Ulm University, EY Foundation).                                                              |
| 2012 – 2013       | <b>Deutschlandstipendium</b> Scholarship (Ulm University, Rotary Club Ulm).                                                            |

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**RESEARCH SEMESTER**

Participant in the **AIM Online Research Community** *4-Dimensional Topology* organized by Miriam Kuzbary, Maggie Miller, Juanita Pinzón-Caicedo and Hannah Schwartz in Spring 2021.

## PREPRINTS

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1. **On the detection of knotted spheres by their traces in high dimensions**, with V. Bais, A. Di Prisa, D. Hartman, C.-S. Hsueh, M. Kegel, A. Merz, M. Pencovitch, A. Ray, D. Santoro and L. Wakelin. [ArXiv:2511.07251](#).

## PUBLISHED AND ACCEPTED PAPERS (IN REVERSE ORDER OF ACCEPTANCE)

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8. **Strongly quasipositive links are concordant to infinitely many strongly quasipositive links**. Accepted for publication in *Proc. Amer. Math. Soc.* [ArXiv:2210.06612](#).
7. **Braid positive surgery diagrams**, with M. Kegel.  
*Glasg. Math. J.* (2026) pp. 1-6. DOI: [10.1017/S0017089526100913](#). [ArXiv:2512.14589](#).
6. **Non-complex cobordisms between quasipositive knots**, with M. Borodzik.  
*J. Math. Pures Appl.* (9) 207:11 (2026). Id/No 103842. [ArXiv:2504.04894](#).
5. **Algorithms in 4-manifold topology**, with S. Bastl, R. Burke, R. Chatterjee, S. Dey, A. Durst, S. Friedl, D. Galvin, A. García Rivas, T. Hirsch, C. Hobohm, C.-S. Hsueh, M. Kegel, F. Kern, S. M. S. Lee, C. Löh, N. Manikandan, L. Mousseau, L. Munser, M. Pencovitch, P. Perras, M. Powell, J. P. Quintanilha, L. Schambeck, D. Suchodoll, M. Tancer, A. Thiele, M. Uschold, S. Veselá, M. Weiß and M. von Wunsch-Rolshoven.  
Accepted for publication in *Algebr. Geom. Topol.*. [ArXiv:2411.08775](#).
4. **Minimal cobordisms between thin and thick torus knots**, with S. Baader, L. Lewark and F. Misev. *Proc. Edinb. Math. Soc., II. Ser.* 69, no. 1 (2026), pp. 177–183. [ArXiv:2405.13719](#).
3. **On the nonorientable four-ball genus of torus knots**, with F. Binns, S. Kang and J. Simone.  
*Algebr. Geom. Topol.* 25-4 (2025), pp. 2209–2251. [ArXiv:2109.09187](#).
2. **3-braid knots with maximal 4-genus**, with S. Baader, L. Lewark and F. Misev.  
*Trans. Amer. Math. Soc. Ser. B* 11 (2024), pp. 600–621. [ArXiv:2303.11918](#).
1. **The upsilon invariant at 1 of 3-braid knots**.  
*Algebr. Geom. Topol.* 23-8 (2023), pp. 3763–3804. [ArXiv:2109.09187](#).

## CONFERENCE PROCEEDINGS (NOT PEER-REVIEWED)

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5. **MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation**, with S. Friedl, J. Licata, S. Tillmann. [Oberwolfach Rep.](#) 21 (2024), no. 3, pp. 2511–2550.
4. **Obstructing ribbon clasp number one**, with D. Celoria, A. Di Prisa, J. Howie, J. Licata, F. Misev, L. Schambeck, L. Wakelin. [Oberwolfach Rep.](#) 21 (2024), no. 3, pp. 2511–2550.
3. **Notions of braid positivity and knot concordance**. [Oberwolfach Rep.](#) 21 (2024), no. 3, pp. 2511–2550.
2. **Strongly quasipositive knots are concordant to infinitely many strongly quasipositive knots**. [Oberwolfach Rep.](#) 20 (2023), no. 1, pp. 172–174.
1. **Concordance of positive braid knots**. [Oberwolfach Rep.](#) 18 (2021), no. 3, pp. 2360–2362.

## INVITED TALKS AT CONFERENCES AND WORKSHOPS

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|          |                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 07/2026* | <a href="#">Young Women in Geometric Topology</a> , Bonn.                                                                                    |
| 06/2026* | Berlin-Brandenburg Workshop VII: Knot theory and its applications, Berlin.                                                                   |
| 10/2025  | <a href="#">Trisections and related topics</a> , Marseille.                                                                                  |
| 11/2024  | <a href="#">IberoSing International Workshop 2024: Low-dimensional Topology &amp; Singularity Theory</a> , Madrid.                           |
| 07/2024  | Mini-symposium <i>Knot theory and low dimensional manifolds</i> , <a href="#">European Congress of Mathematics</a> , Seville.                |
| 04/2024  | <a href="#">NRW Topology Meeting</a> , Wuppertal.                                                                                            |
| 04/2024  | <a href="#">Manifolds and groups in Bologna II</a> , Bologna.                                                                                |
| 09/2023  | <a href="#">Leaning into Topology Workshop</a> , Pisa.                                                                                       |
| 09/2023  | <a href="#">Swiss Knots</a> , Regensburg.                                                                                                    |
| 06/2023  | Mini Workshop, Bern.                                                                                                                         |
| 03/2023  | <a href="#">The low dimensional workshop</a> , Budapest,  . |
| 11/2022  | <a href="#">Paroles aux jeunes chercheuses et chercheurs en géométrie et dynamique</a> , Orsay.                                              |
| 09/2022  | Mini-symposium <i>Algebra and Low-Dimensional Topology</i> , <a href="#">DMV Annual Meeting 2022</a> , Berlin.                               |
| 06/2022  | <a href="#">Surfaces in 4-manifolds</a> , Le Croisic.                                                                                        |

## INVITED TALKS AT SEMINARS

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|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02/2026 | <a href="#">Geometry and Topology Seminar</a> , Glasgow.                                                                                                          |
| 01/2026 | <a href="#">Geometry Seminar</a> , MPI MiS Leipzig.                                                                                                               |
| 10/2025 | <a href="#">Algebra and Geometry Seminar</a> , Neuchâtel.                                                                                                         |
| 07/2025 | <a href="#">Oberseminar Global Analysis and Geometry</a> , Regensburg.                                                                                            |
| 06/2025 | <a href="#">Séminaire Géométrie des Espaces Singuliers</a> , Lille.                                                                                               |
| 05/2025 | <a href="#">Geometry seminar</a> , ETH Zurich.                                                                                                                    |
| 12/2024 | <a href="#">Oberseminar Topologie</a> , University of Bonn.                                                                                                       |
| 09/2024 | <a href="#">Séminaire de Géométrie, Algèbre, Dynamique et Topologie</a> , Dijon.                                                                                  |
| 05/2024 | <a href="#">Séminaire de Géométrie et Topologie de Marseille</a> , Marseille.                                                                                     |
| 05/2024 | <a href="#">Oberseminar Topologie</a> , Münster.                                                                                                                  |
| 04/2024 | <a href="#">Geometry &amp; Topology Seminar</a> , Glasgow.                                                                                                        |
| 03/2024 | <a href="#">MPIM Oberseminar</a> , MPIM Bonn.                                                                                                                     |
| 11/2023 | <a href="#">MPIM Topology Seminar</a> , MPIM Bonn, short talk.                                                                                                    |
| 05/2023 | <a href="#">Low-dimensional topology seminar</a> , MPIM Bonn.                                                                                                     |
| 12/2022 | <a href="#">Geometry and Topology Seminar</a> , CIRGET, Montréal, online,  . |
| 06/2022 | <a href="#">LKS-Seminar</a> , Regensburg.                                                                                                                         |
| 05/2022 | <a href="#">Geometry &amp; Topology Seminar</a> , Caltech, online.                                                                                                |
| 05/2022 | <a href="#">Oberseminar Geometrie</a> , Fribourg.                                                                                                                 |
| 02/2022 | <a href="#">Knot Online Seminar (K-OS)</a> , online,  .                        |
| 04/2021 | AIM Online Research Community <a href="#">4-Dimensional Topology</a> , lightning talk.                                                                            |

## INVITED TALKS AT GRADUATE COLLOQUIA

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|---------|---------------------------------------------------------------------------------|
| 12/2023 | <a href="#">Pitt AWM Student Seminar</a> , Pittsburgh, online.                  |
| 10/2023 | <a href="#">Geometry Graduate Colloquium</a> , ETH Zurich.                      |
| 10/2022 | <a href="#">Bern–Fribourg Graduate Seminar</a> , Bern.                          |
| 05/2022 | <a href="#">Baby Geometri Seminar</a> , Pisa.                                   |
| 04/2022 | <a href="#">Bernoullis Tafelrunde</a> , Basel.                                  |
| 11/2021 | <a href="#">Zurich Graduate Colloquium</a> , Zurich.                            |
| 04/2021 | <a href="#">Geometric Topology Grad and Postdoc Seminar (GT GAPS)</a> , online. |

## CONTRIBUTED TALKS AT CONFERENCES AND WORKSHOPS

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|         |                                                                                                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2024 | <a href="#">MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation</a> , MFO Oberwolfach, short talk. |
| 02/2024 | <a href="#">BTW4: Knot Floer Homology</a> , Besse.                                                                                                          |
| 02/2023 | <a href="#">Winter Braids XII</a> , Tours.                                                                                                                  |
| 01/2023 | <a href="#">Low dimensional topology</a> , MFO Oberwolfach.                                                                                                 |
| 05/2022 | <a href="#">Swiss-Mathematical-Society Doctoral Day 2022</a> , Fribourg.                                                                                    |
| 04/2022 | <a href="#">BTW2: Sliceness, exotic pairs, and quantum invariants</a> , Aussois.                                                                            |
| 09/2021 | <a href="#">Invariants and Structures in Low-Dimensional Topology</a> , MFO Oberwolfach, short talk.                                                        |

## OTHER CONFERENCES AND WORKSHOPS ATTENDED

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|          |                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| 07/2026* | <a href="#">Low-dimensional Topology</a> , MFO Oberwolfach.                                                                          |
| 06/2026* | <a href="#">Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions</a> , Budapest.                          |
| 12/2025  | <a href="#">Workshop on surfaces in 4-manifolds</a> , Regensburg.                                                                    |
| 10/2025  | <a href="#">36. NRW Topology Meeting</a> , Bonn.                                                                                     |
| 07/2025  | <a href="#">Categorification in Low Dimensional Topology</a> , Bochum.                                                               |
| 06/2025  | <a href="#">Swiss Knots</a> , Geneva.                                                                                                |
| 09/2024  | <a href="#">Workshop on 4-manifolds and algorithms</a> , Regensburg.                                                                 |
| 02/2024  | <a href="#">Winter Braids XIII</a> , Montpellier.                                                                                    |
| 09/2023  | <a href="#">GRAZP: Groups and Rigidity Around the Zimmer Program</a> , Ventotene.                                                    |
| 07/2023  | <a href="#">Gauge Theory and Topology: in Celebration of Peter Kronheimer's 60th Birthday</a> , Oxford.                              |
| 01/2023  | <a href="#">Winter school on singularities and low-dimensional topology</a> , Budapest.                                              |
| 08/2022  | <a href="#">Frontiers in Geometry and Topology Summer School</a> , Trieste.                                                          |
| 04/2022  | <a href="#">Braids in Low-Dimensional Topology</a> , ICERM, Providence.                                                              |
| 08/2021  | <a href="#">Perspectives on quantum link homology theories</a> , Universität Regensburg.                                             |
| 07/2021  | <a href="#">Tech Topology Summer School</a> , Georgia Institute of Technology, online.                                               |
| 07/2021  | <a href="#">Swiss Knots</a> , Fribourg.                                                                                              |
| 06/2021  | <a href="#">NCNGT: Nearly Carbon Neutral Geometric Topology Conference</a> , online.                                                 |
| 06/2020  | <a href="#">NCNGT: Nearly Carbon Neutral Geometric Topology Conference</a> , online.                                                 |
| 02/2020  | <a href="#">Winter Braids X</a> , Pisa.                                                                                              |
| 10/2019  | <a href="#">Regional workshop on foliations and 3-manifolds</a> , Universität Regensburg.                                            |
| 10/2019  | <a href="#">Low-dimensional topology workshop</a> , Universität Regensburg.                                                          |
| 09/2019  | <a href="#">Workshop on 4-manifolds</a> , Max Planck Institute for Mathematics Bonn.                                                 |
| 09/2019  | <a href="#">Of coarse! Quasi-isometries and groups: rigidity and classification</a> , Ventotene.                                     |
| 07/2019  | <a href="#">Swiss Knots</a> , ETH Zurich.                                                                                            |
| 10/2017  | <a href="#">27th annual Route 81 conference on commutative algebra and algebraic geometry</a> , Cornell University, Ithaca, NY, USA. |

\* means planned.

## SUPERVISION

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- 02/2025 – 10/2025    **Master thesis** by Chiara Donatone at Università degli Studi di Milano.  
Thesis title: *On RBG links and knot traces*.  
Co-supervision with Diego Matessi.  
Will continue as PhD student at the University of Glasgow.
- 06/2024 – 06/2025    **Master thesis** by Frieda Kern at the University of Bonn.  
Thesis title: *Linking number obstructions to satellite homomorphisms*.  
Co-supervision with Peter Teichner.  
Continues as [PhD student](#) at the University of Glasgow.

## TEACHING

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- Winter 2024/25    **University of Bonn/MPIM Bonn**  
Lecturer for [Topics in 3-manifold topology](#) (Master in Mathematics).
- Spring 2022    **ETH Zurich**  
Course organizer for **Analysis II** (Bachelor in Civil, Geospatial, or Environmental Engineering).
- Fall 2021    Course organizer for **Analysis I** (Bachelor in Civil, Geospatial, or Environmental Engineering).
- Spring 2021    Course organizer for [Topologie](#) (Bachelor in Mathematics).
- Fall 2020    Course organizer for [Lineare Algebra I](#) (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
- Spring 2020    Course organizer for [Algebraic Topology II](#) (Bachelor or Master in Mathematics).
- Fall 2019    Teaching assistant for [Complex Analysis](#) (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
- Spring 2018    **Syracuse University**  
Teaching assistant for **Calculus III**.
- Fall 2017    Teaching assistant for **Calculus III**.
- Summer 2019    **Ulm University**  
Instructor for **Linear Algebra Review Course** (Bachelor in Computer Science and Engineering).
- Winter 2013/14 – Summer 2019    Student teaching assistant for **Analysis I-III** and **Linear Algebra I-II**.

## TEACHER TRAINING

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- Winter 2024/25    Participation in the [Teaching Mentoring Program](#), Hausdorff School for Mathematics in Bonn.  
Mentors: Viktoriya Ozornova and Arunima Ray (both MPIM Bonn).
- 09/2021    Certificate **Learning to Teach**.  
Program for doctoral teaching assistants, ETH Zurich.  
Teaching project: *How to teach writing proofs—classroom assessment techniques in mathematics classes*.

## OUTREACH ACTIVITIES

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|         |                                                                                                                                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2026 | Co-organizer of <b>GROW@Dresden 2026</b> (Graduate Research Opportunities for Women at Dresden 2026), March 26-27.<br>Joint with Anita Behme, Türkü Özlüm Çelik, Heather Harrington, Oana Padurariu, Andreas Thom. |
| 03/2025 | Invited panelist for panel discussion “ <i>What is mathematics research like?</i> ” at <b>GROW@Frankfurt 2025</b> .                                                                                                |
| 12/2024 | Invited speaker at <b>Tea Time with Women in Mathematics</b> , Bonn.                                                                                                                                               |
| 04/2024 | Volunteer at the mentoring lunch at <b>GROW@Bonn 2024</b> .                                                                                                                                                        |
| 03/2023 | <i>Work as a PhD Student</i> , invited short talk, <b>MindPhair 2023</b> , ETH Zurich (job-fair for mathematicians, physicists and computational scientists).                                                      |
| 03/2022 | <i>Work as a PhD Student</i> , invited short talk, <b>MindPhair 2022</b> , ETH Zurich.                                                                                                                             |

## CONFERENCE AND SEMINAR ORGANIZATION

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| 02/2026 –         | <b>ArXiv seminar</b> , University of Glasgow.                                                                                                                                                                                                   |
| 10/2023 – 11/2025 | <b>Low-dimensional topology seminar</b> , MPIM Bonn.                                                                                                                                                                                            |
| 09/2024           | <b>MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation</b> , Mathematisches Forschungsinstitut Oberwolfach, Sept 22–27.<br>Joint with Stefan Friedl, Joan Licata and Stephan Tillmann. |
| 09/2020 – 07/2023 | <b>Geometry Graduate Colloquium</b> , ETH Zurich, funded by the Zurich Graduate School in Mathematics.                                                                                                                                          |

## OTHER ACADEMIC SERVICE AND ACTIVITIES

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <b>MPIM Bonn</b>                                                                                                                                 |
| 09/2024           | Member of the Postdoc Representative Selection Committee.                                                                                        |
|                   | <b>ETH Zurich</b>                                                                                                                                |
| 09/2021 – 07/2023 | Co-organizer of get-togethers for women in math at ETH Zurich and the University of Zurich, funded by the Zurich Graduate School in Mathematics. |
| 08/2021 – 07/2023 | Elected member of the Nachdiplom Lectures Committee at the Department of Mathematics.                                                            |
| 05/2020 – 05/2023 | Member of the Board of <b>VMM</b> (Association of the Mid-level Academic Faculty at the Department of Mathematics).                              |
| 10/2022           | Co-organizer of the first D-MATH (Post)Doctoral Welcome Day.                                                                                     |
| 05/2021 – 05/2022 | Mid-level faculty representative in the Department Conference.                                                                                   |
|                   | <b>Ulm University</b>                                                                                                                            |
| 10/2016 – 07/2017 | Member of the Mathematics Student Council (Fachschaft Mathematik) at the Department of Mathematics and Economics.                                |

## REFEREING AND REVIEWING

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|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 2024 – present | <b>Referee</b> and <b>quick opinions</b> for <i>Algebr. Geom. Topol.</i> , <i>Bull. Lond. Math. Soc.</i> and <i>Pac. J. Math.</i> |
| 2024 – present | <b>Reviewer</b> for <i>zbMATH Open</i> .                                                                                          |

## REFERENCES

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[Peter Feller](#), University of Neuchâtel,  
[Matthew Hedden](#), Michigan State University,  
[Lukas Lewark](#), ETH Zurich,  
[Viktoriya Ozornova](#), MPIM Bonn (Teaching),  
[Arunima Ray](#), University of Melbourne,

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[viktoriya.ozornova@mpim-bonn.mpg.de](mailto:viktoriya.ozornova@mpim-bonn.mpg.de).  
[aru.ray@unimelb.edu.au](mailto:aru.ray@unimelb.edu.au).